

DSC 140B

Representation Learning

Lecture 03 | Part 1

News

News

- ▶ Cheat sheet allowed on quizzes.
- ▶ FinAid survey on Gradescope.
 - ▶ If you're here in person (and participate in the Live Q&A), you don't need to do this.
- ▶ Practice problems now on dsc140b.com

DSC 140B

Representation Learning

Lecture 03 | Part 2

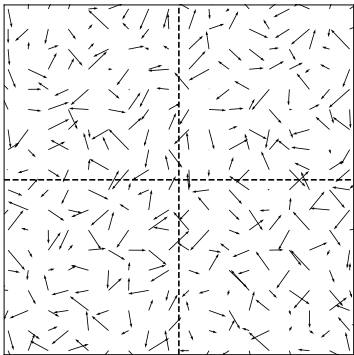
Functions of a Vector

Transformations

- ▶ A **transformation** $\vec{f}$ is a function that takes in a vector, and returns a vector *of the same dimensionality*.
- ▶ That is, $\vec{f} : \mathbb{R}^d \rightarrow \mathbb{R}^d$.

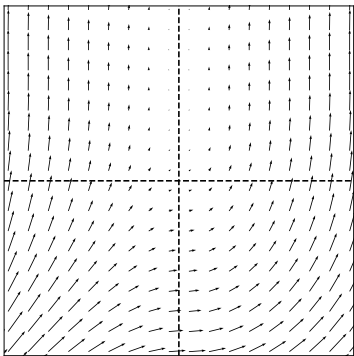
Arbitrary Transformations

- Arbitrary transformations can be quite complex.



Arbitrary Transformations

- Arbitrary transformations can be quite complex.



Linear Transformations

- ▶ Luckily, we often¹ work with simpler, **linear transformations**.
- ▶ A transformation f is **linear** if:

$$\vec{f}(\alpha\vec{x} + \beta\vec{y}) = \alpha\vec{f}(\vec{x}) + \beta\vec{f}(\vec{y})$$

¹Sometimes just to make the math tractable!

Key Fact

- ▶ Linear functions are determined **entirely** by what they do on the basis vectors.
- ▶ I.e., to tell you what f does, I only need to tell you $\vec{f}(\hat{e}^{(1)})$ and $\vec{f}(\hat{e}^{(2)})$.
- ▶ This makes the math easy!

Linear Algebra

- ▶ This is the key idea behind **linear** algebra.
- ▶ Linear algebra studies the properties of **linear** transformations.
- ▶ Non-linear transformations are **so complicated** that we can say relatively little about them.

A photograph of a formal garden, likely the gardens of Stourhead in England. The garden features a central, wide, rectangular lawn path that leads from the foreground towards the background. On either side of the path, there are symmetrical, rectangular hedges that form a series of terraces. The hedges are filled with various plants, including tall grasses, shrubs, and small flowers. The background is a dense, lush forest with tall trees and thick foliage. The overall scene is a well-maintained, symmetrical landscape.

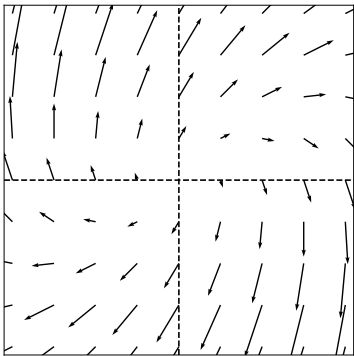
Arbitrary
Transformations

Linear
Transformations

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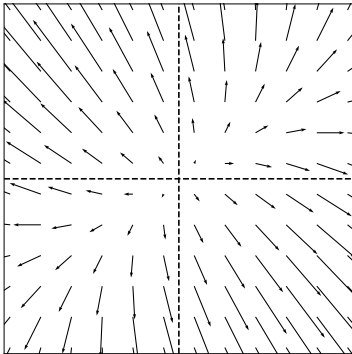
Example Linear Transformation

► $\vec{f}(\vec{x}) = (x_1 + 3x_2, -3x_1 + 5x_2)^T$



Another Example Linear Transformation

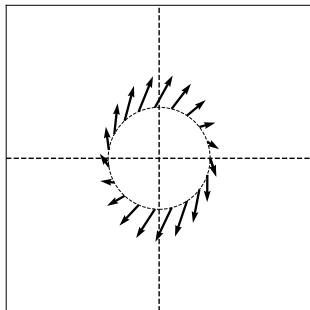
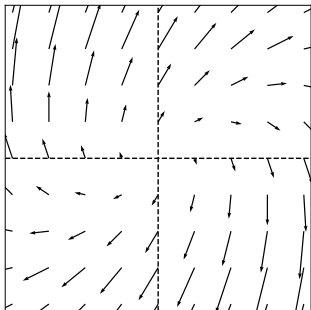
► $\vec{f}(\vec{x}) = (2x_1 - x_2, -x_1 + 3x_2)^T$

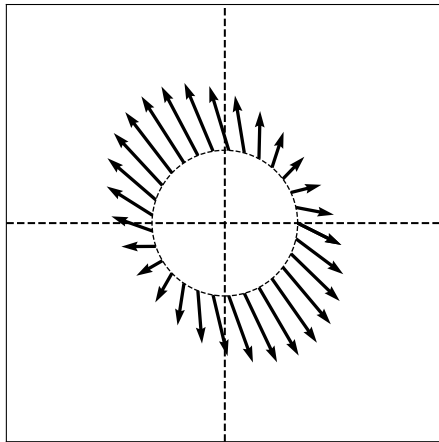
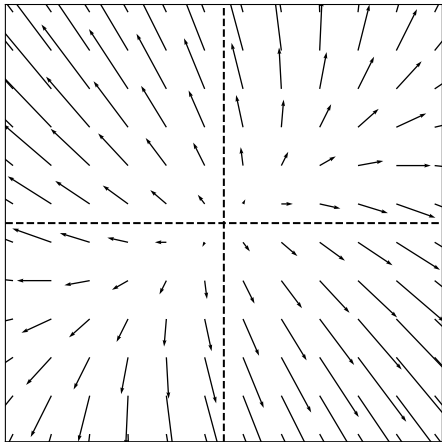


Note

- Because of linearity, along any given direction $\vec{f}$ changes only in scale.

$$\vec{f}(\lambda \hat{x}) = \lambda \vec{f}(\hat{x})$$





Linear Transformations and Bases

- We have been writing transformations in coordinate form. For example:

$$\begin{aligned}\vec{f}(\vec{x}) &= (x_1 + x_2, x_1 - x_2)^T \\ &= (x_1 + x_2)\hat{e}^{(1)} + (x_1 - x_2)\hat{e}^{(2)}\end{aligned}$$

- If we use a different basis, the formula for $\vec{f}$ **changes**:

$$\begin{aligned}[\vec{f}(\vec{x})]_{\mathcal{U}} &= (?, ?)^T \\ &= [?]\hat{u}^{(1)} + [?]\hat{u}^{(2)}\end{aligned}$$

Linear Transformations and Bases

- We know that if $\vec{x} = x_1\hat{e}^{(1)} + x_2\hat{e}^{(2)}$, then:

$$\vec{f}(\vec{x}) = (x_1 + x_2)\hat{e}^{(1)} + (x_1 - x_2)\hat{e}^{(2)}$$

- Now: if $\vec{x} = z_1\hat{u}^{(1)} + z_2\hat{u}^{(2)}$, what is:

$$\vec{f}(\vec{x}) = ?\hat{u}^{(1)} + ?\hat{u}^{(2)}$$

Key Fact

- If we use linearity:

$$\begin{aligned}f(\vec{x}) &= f(z_1 \hat{u}^{(1)} + z_2 \hat{u}^{(2)}) \\ &= z_1 f(\hat{u}^{(1)}) + z_2 f(\hat{u}^{(2)})\end{aligned}$$

- **Strategy:** to write $\vec{f}$ in the $\mathcal{U}$ basis, we just need to know what $\vec{f}$ does to $\hat{u}^{(1)}$ and $\hat{u}^{(2)}$.

Example

► Let:

► $\vec{f}(\vec{x}) = (x_1 + x_2, x_1 - x_2)^T$
► $\hat{u}^{(1)} = \frac{1}{\sqrt{2}}(1, 1)^T$ and $\hat{u}^{(2)} = \frac{1}{\sqrt{2}}(-1, 1)^T$.

► Then:

$$\vec{f}(\hat{u}^{(1)}) = \vec{f}\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^T = (\sqrt{2}, 0)^T = \sqrt{2}\hat{e}^{(1)}$$

$$\vec{f}(\hat{u}^{(2)}) = \vec{f}\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^T = (0, -\sqrt{2})^T = -\sqrt{2}\hat{e}^{(2)}$$

► **But** we want $\vec{f}(\hat{u}^{(1)})$ and $\vec{f}(\hat{u}^{(2)})$ in terms of $\hat{u}^{(1)}$ and $\hat{u}^{(2)}$.

Example (Cont.)

- ▶ We have: $f(\hat{u}^{(1)}) = \sqrt{2}\hat{e}^{(1)}$ and $f(\hat{u}^{(2)}) = -\sqrt{2}\hat{e}^{(2)}$.
- ▶ To write $\vec{f}(\hat{u}^{(1)})$ in terms of $\hat{u}^{(1)}$ and $\hat{u}^{(2)}$, compute:

$$\begin{aligned} f(\hat{u}^{(1)}) &= (f(\hat{u}^{(1)}) \cdot \hat{u}^{(1)})\hat{u}^{(1)} + (f(\hat{u}^{(1)}) \cdot \hat{u}^{(2)})\hat{u}^{(2)} \\ &= \\ &= \end{aligned}$$

Example (Cont.)

- ▶ We have: $f(\hat{u}^{(1)}) = \sqrt{2}\hat{e}^{(1)}$ and $f(\hat{u}^{(2)}) = -\sqrt{2}\hat{e}^{(2)}$.
- ▶ To write $\vec{f}(\hat{u}^{(1)})$ in terms of $\hat{u}^{(1)}$ and $\hat{u}^{(2)}$, compute:

$$\begin{aligned} f(\hat{u}^{(1)}) &= (f(\hat{u}^{(1)}) \cdot \hat{u}^{(1)})\hat{u}^{(1)} + (f(\hat{u}^{(1)}) \cdot \hat{u}^{(2)})\hat{u}^{(2)} \\ &= \left((\sqrt{2}, 0) \cdot \frac{1}{\sqrt{2}}(1, 1) \right) \hat{u}^{(1)} + \left((\sqrt{2}, 0) \cdot \frac{1}{\sqrt{2}}(-1, 1) \right) \hat{u}^{(2)} \\ &= \end{aligned}$$

Example (Cont.)

- ▶ We have: $f(\hat{u}^{(1)}) = \sqrt{2}\hat{e}^{(1)}$ and $f(\hat{u}^{(2)}) = -\sqrt{2}\hat{e}^{(2)}$.
- ▶ To write $\vec{f}(\hat{u}^{(1)})$ in terms of $\hat{u}^{(1)}$ and $\hat{u}^{(2)}$, compute:

$$\begin{aligned}f(\hat{u}^{(1)}) &= (f(\hat{u}^{(1)}) \cdot \hat{u}^{(1)})\hat{u}^{(1)} + (f(\hat{u}^{(1)}) \cdot \hat{u}^{(2)})\hat{u}^{(2)} \\&= \left((\sqrt{2}, 0) \cdot \frac{1}{\sqrt{2}}(1, 1) \right) \hat{u}^{(1)} + \left((\sqrt{2}, 0) \cdot \frac{1}{\sqrt{2}}(-1, 1) \right) \hat{u}^{(2)} \\&= (1)\hat{u}^{(1)} + (-1)\hat{u}^{(2)} = \hat{u}^{(1)} - \hat{u}^{(2)}\end{aligned}$$

Example (Cont.)

- Similarly, for $\vec{f}(\hat{u}^{(2)})$:

$$\begin{aligned} f(\hat{u}^{(2)}) &= (f(\hat{u}^{(2)}) \cdot \hat{u}^{(1)})\hat{u}^{(1)} + (f(\hat{u}^{(2)}) \cdot \hat{u}^{(2)})\hat{u}^{(2)} \\ &= \left((0, -\sqrt{2}) \cdot \frac{1}{\sqrt{2}}(1, 1) \right) \hat{u}^{(1)} + \left((0, -\sqrt{2}) \cdot \frac{1}{\sqrt{2}}(-1, 1) \right) \hat{u}^{(2)} \\ &= (-1)\hat{u}^{(1)} + (-1)\hat{u}^{(2)} = -\hat{u}^{(1)} - \hat{u}^{(2)} \end{aligned}$$

Solution

- Putting it all together:

$$\begin{aligned}f(\vec{X}) &= f(z_1 \hat{u}^{(1)} + z_2 \hat{u}^{(2)}) \\&= z_1 f(\hat{u}^{(1)}) + z_2 f(\hat{u}^{(2)}) \\&= z_1 (\hat{u}^{(1)} - \hat{u}^{(2)}) + z_2 (-\hat{u}^{(1)} - \hat{u}^{(2)}) \\&= (z_1 - z_2) \hat{u}^{(1)} + (-z_1 - z_2) \hat{u}^{(2)}\end{aligned}$$

- Or, in coordinate form:

$$[f(\vec{X})]_{\mathcal{U}} = (z_1 - z_2, -z_1 - z_2)^T$$

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Representation Learning

Lecture 03 | Part 3

Matrices

Matrices?

- ▶ I thought this week was supposed to be about linear algebra... Where are the matrices?

Matrices?

- ▶ I thought this week was supposed to be about linear algebra... Where are the matrices?
- ▶ What is a matrix, anyways?

What is a matrix?

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Recall: Linear Transformations

- ▶ A **transformation** $\vec{f}(\vec{x})$ is a function which takes a vector as input and returns a vector of the same dimensionality.
- ▶ A transformation $\vec{f}$ is **linear** if

$$\vec{f}(\alpha \vec{u} + \beta \vec{v}) = \alpha \vec{f}(\vec{u}) + \beta \vec{f}(\vec{v})$$

Recall: Linear Transformations

- ▶ **Key** consequence of **linearity**: to compute $\vec{f}(\vec{x})$, only need to know what $\vec{f}$ does to basis vectors.
- ▶ Example:

$$\vec{x} = 3\hat{e}^{(1)} - 4\hat{e}^{(2)} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$

$$\vec{f}(\hat{e}^{(1)}) = -\hat{e}^{(1)} + 3\hat{e}^{(2)}$$

$$\vec{f}(\hat{e}^{(2)}) = 2\hat{e}^{(1)}$$

$$\vec{f}(\vec{x}) =$$

Matrices

- **Idea:** Since $\vec{f}$ is defined by what it does to basis, place $\vec{f}(\hat{e}^{(1)})$, $\vec{f}(\hat{e}^{(2)})$, ... into a table as columns
- This is the **matrix** representing² $\vec{f}$

$$\begin{aligned}\vec{f}(\hat{e}^{(1)}) &= -\hat{e}^{(1)} + 3\hat{e}^{(2)} = \begin{pmatrix} -1 \\ 3 \end{pmatrix} \\ \vec{f}(\hat{e}^{(2)}) &= 2\hat{e}^{(1)} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}\end{aligned}\qquad \begin{pmatrix} -1 & 2 \\ 3 & 0 \end{pmatrix}$$

²with respect to the standard basis $\hat{e}^{(1)}, \hat{e}^{(2)}$

Example

Write the matrix representing $\vec{f}$ with respect to the standard basis, given:

$$\vec{f}(\hat{e}^{(1)}) = (1, 4, 7)^T$$

$$\vec{f}(\hat{e}^{(2)}) = (2, 5, 8)^T$$

$$\vec{f}(\hat{e}^{(3)}) = (3, 6, 9)^T$$

Exercise

Suppose $\vec{f}$ has the matrix below:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Let $\vec{x} = (-2, 1, 3)^T$. What is $\vec{f}(\vec{x})$?

- ▶ A) $(3, 12, 21)^T$
- ▶ B) $(-2, 1, 3)^T$
- ▶ C) $(6, 15, 24)^T$
- ▶ D) $(9, 15, 21)^T$

Main Idea

A square ($n \times n$) matrix can be interpreted as a compact representation of a linear transformation $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

What is matrix multiplication?

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} \\ \\ \end{pmatrix}$$

A low-level definition

$$(A\vec{x})_i = \sum_{j=1}^n A_{ij}x_j$$

A low-level interpretation

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} = -2 \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} + 1 \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} + 3 \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}$$

In general...

$$\begin{pmatrix} \uparrow & \uparrow & \uparrow \\ \vec{a}^{(1)} & \vec{a}^{(2)} & \vec{a}^{(3)} \\ \downarrow & \downarrow & \downarrow \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1 \vec{a}^{(1)} + x_2 \vec{a}^{(2)} + x_3 \vec{a}^{(3)}$$

Matrix Multiplication

$$\vec{x} = x_1 \hat{e}^{(1)} + x_2 \hat{e}^{(2)} + x_3 \hat{e}^{(3)} = (x_1, x_2, x_3)^T$$
$$\vec{f}(\vec{x}) = x_1 \vec{f}(\hat{e}^{(1)}) + x_2 \vec{f}(\hat{e}^{(2)}) + x_3 \vec{f}(\hat{e}^{(3)})$$

$$A = \begin{pmatrix} \begin{matrix} \uparrow \\ \vec{f}(\hat{e}^{(1)}) \\ \downarrow \end{matrix} & \begin{matrix} \uparrow \\ \vec{f}(\hat{e}^{(2)}) \\ \downarrow \end{matrix} & \begin{matrix} \uparrow \\ \vec{f}(\hat{e}^{(3)}) \\ \downarrow \end{matrix} \end{pmatrix}$$
$$A\vec{x} = \begin{pmatrix} \begin{matrix} \uparrow \\ \vec{f}(\hat{e}^{(1)}) \\ \downarrow \end{matrix} & \begin{matrix} \uparrow \\ \vec{f}(\hat{e}^{(2)}) \\ \downarrow \end{matrix} & \begin{matrix} \uparrow \\ \vec{f}(\hat{e}^{(3)}) \\ \downarrow \end{matrix} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$
$$= x_1 \vec{f}(\hat{e}^{(1)}) + x_2 \vec{f}(\hat{e}^{(2)}) + x_3 \vec{f}(\hat{e}^{(3)})$$

Matrix Multiplication

- ▶ Matrix A represents a linear transformation $\vec{f}$
 - ▶ With respect to the standard basis
 - ▶ If we use a different basis, the matrix changes!
- ▶ Matrix multiplication $A\vec{x}$ **evaluates** $\vec{f}(\vec{x})$

What are they, *really*?

- ▶ Matrices are sometimes just tables of numbers.
- ▶ But they often have a deeper meaning.

Main Idea

A square ($n \times n$) matrix can be interpreted as a compact representation of a linear transformation $\vec{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

What's more, if A represents $\vec{f}$, then $A\vec{x} = \vec{f}(\vec{x})$; that is, multiplying by A is the same as evaluating $\vec{f}$.

Example

$$\vec{x} = 3\hat{e}^{(1)} - 4\hat{e}^{(2)} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$

$$A =$$

$$\vec{f}(\hat{e}^{(1)}) = -\hat{e}^{(1)} + 3\hat{e}^{(2)}$$

$$\vec{f}(\hat{e}^{(2)}) = 2\hat{e}^{(1)}$$

$$\vec{f}(\vec{x}) =$$

$$A\vec{x} =$$

Note

- ▶ All of this works because we assumed $\vec{f}$ is **linear**.
- ▶ If it isn't, evaluating $\vec{f}$ isn't so simple.

Note

- ▶ All of this works because we assumed $\vec{f}$ is **linear**.
- ▶ If it isn't, evaluating $\vec{f}$ isn't so simple.
- ▶ Linear algebra = simple!

Matrices in Other Bases

- The matrix of a linear transformation wrt the **standard basis**:

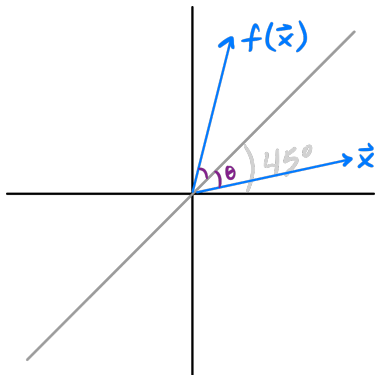
$$\begin{pmatrix} \uparrow & \uparrow & \uparrow & \\ \vec{f}(\hat{e}^{(1)}) & \vec{f}(\hat{e}^{(2)}) & \cdots & \vec{f}(\hat{e}^{(d)}) \\ \downarrow & \downarrow & \downarrow & \end{pmatrix}$$

- With respect to basis $\mathcal{U}$:

$$\begin{pmatrix} \uparrow & \uparrow & \uparrow & \\ [\vec{f}(\hat{u}^{(1)})]_{\mathcal{U}} & [\vec{f}(\hat{u}^{(2)})]_{\mathcal{U}} & \cdots & [\vec{f}(\hat{u}^{(d)})]_{\mathcal{U}} \\ \downarrow & \downarrow & \downarrow & \end{pmatrix}$$

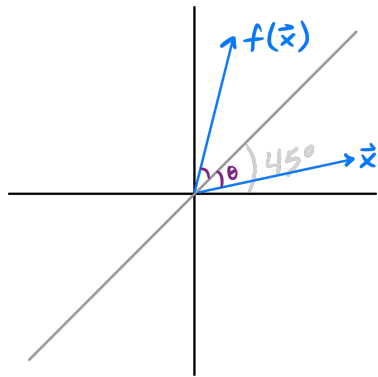
Example

- Consider the transformation $\vec{f}$ which “mirrors” a vector over the line of 45° .



- What is its matrix in the standard basis?

Example



- ▶ Let $\hat{u}^{(1)} = \frac{1}{\sqrt{2}}(1, 1)^T$
- ▶ Let $\hat{u}^{(2)} = \frac{1}{\sqrt{2}}(-1, 1)^T$
- ▶ What is $[\vec{f}(\hat{u}^{(1)})]_{\mathcal{U}}$?
- ▶ $[\vec{f}(\hat{u}^{(2)})]_{\mathcal{U}}$?
- ▶ What is the matrix?

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Representation Learning

Lecture 03 | Part 4

The Spectral Theorem

Eigenvectors

- Let A be an $n \times n$ matrix. An **eigenvector** of A with **eigenvalue** λ is a nonzero vector $\vec{v}$ such that $A\vec{v} = \lambda\vec{v}$.

Eigenvectors (of Linear Transformations)

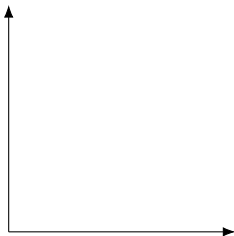
- Let $\vec{f}$ be a linear transformation. An **eigenvector** of $\vec{f}$ with **eigenvalue** λ is a nonzero vector $\vec{v}$ such that $f(\vec{v}) = \lambda\vec{v}$.

Importance

- ▶ We will see why eigenvectors are important in the next part.
- ▶ For now: what are they?

Geometric Interpretation

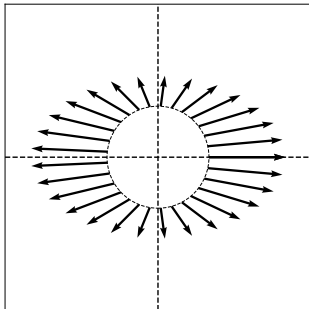
- ▶ When $\vec{f}$ is applied to one of its eigenvectors, $\vec{f}$ simply scales it.
 - ▶ Possibly by a negative amount.



Exercise

Draw as many (linearly independent) eigenvectors as you can:

$$A = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$$



Finding Eigenvectors

- ▶ We typically compute the eigenvectors of a matrix with a computer.
- ▶ But it can help our understanding to find them “graphically”.

Procedure

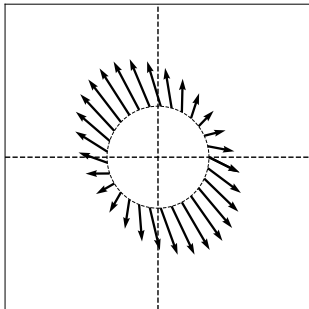
Given a matrix A (or transformation $\vec{f}$), to find an eigenvector “graphically”.

1. Think about (or draw) the output of $\vec{f}$ for a handful of unit vector inputs.
 - ▶ Linear transformations are continuous so you can “interpolate”.
2. Find place(s) where the input vector and the output vector are parallel.

Exercise

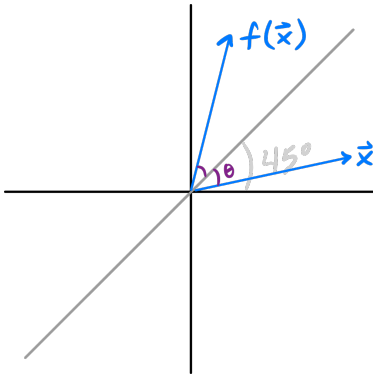
Draw as many (linearly independent) eigenvectors as you can:

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$$



Exercise

Consider the linear transformation which mirrors its input over the line of 45° . Give two orthogonal eigenvectors of the transformation.



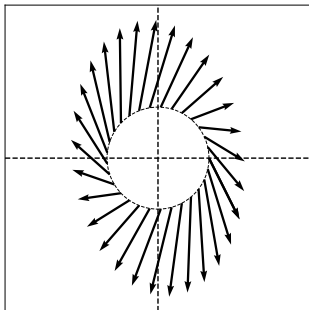
Alternate Procedure: Guess and Check

1. Guess a vector $\vec{x}$.
2. Check that $\vec{f}(\vec{x}) = \lambda\vec{x}$.

Exercise

Draw as many (linearly independent) eigenvectors as you can:

$$A = \begin{pmatrix} 5 & 5 \\ -10 & 12 \end{pmatrix}$$



Caution!

- ▶ Not all matrices have even one eigenvector!³
- ▶ When does a matrix have multiple (linearly independent) eigenvectors?

³That is, with a *real-valued* eigenvalue.

Symmetric Matrices

- Recall: a matrix A is **symmetric** if $A^T = A$.

The Spectral Theorem⁴

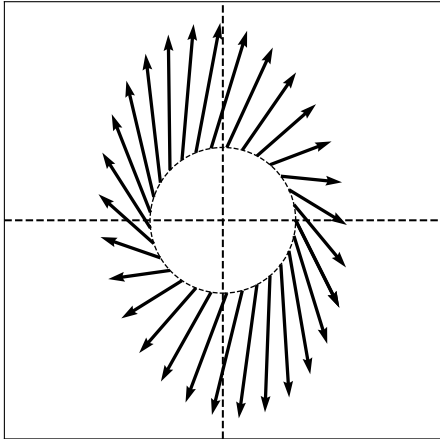
- ▶ **Theorem:** Let A be an $n \times n$ *symmetric* matrix. Then there exist n eigenvectors of A which are all mutually orthogonal.

⁴for symmetric matrices

What?

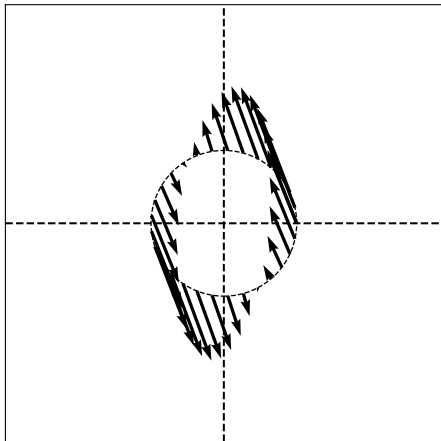
- ▶ What does the spectral theorem mean?
- ▶ What is an eigenvector, really?
- ▶ Why are they useful?

Example Linear Transformation



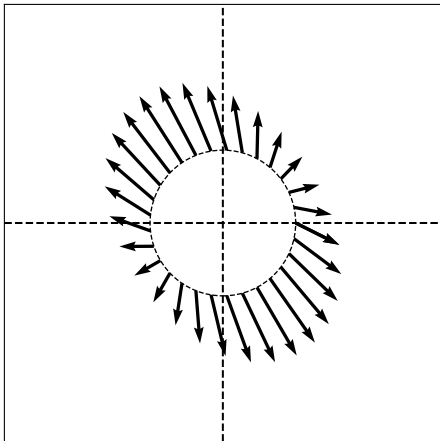
$$A = \begin{pmatrix} 5 & 5 \\ -10 & 12 \end{pmatrix}$$

Example Linear Transformation



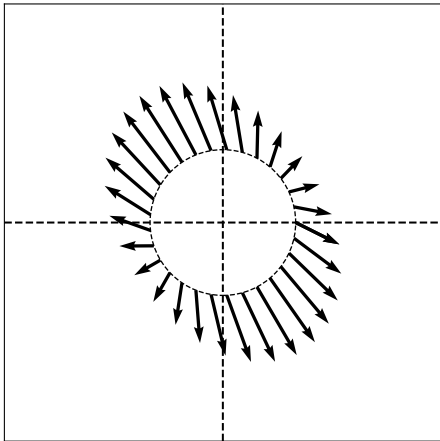
$$A = \begin{pmatrix} -2 & -1 \\ -5 & 3 \end{pmatrix}$$

Example Symmetric Linear Transformation



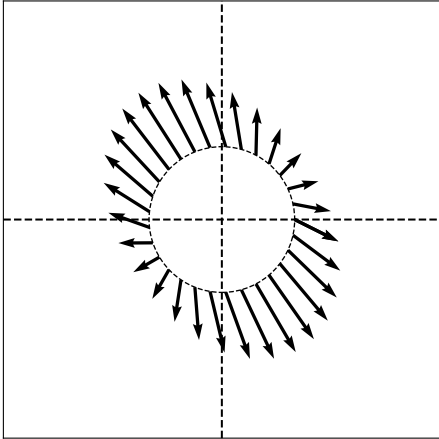
$$A = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$$

Observation #1



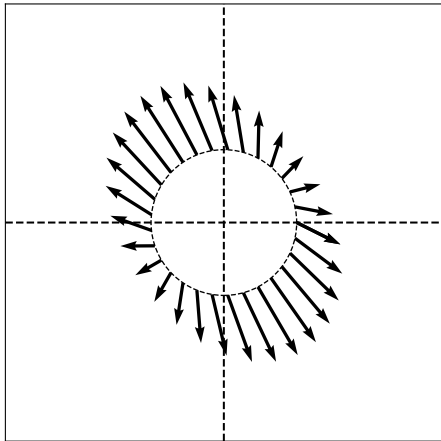
- Symmetric linear transformations have **axes of symmetry**.

Observation #2



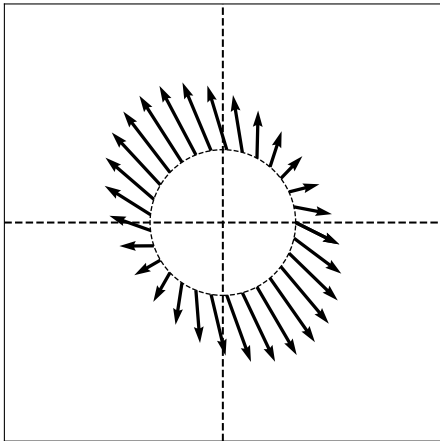
- The axes of symmetry are **orthogonal** to one another.

Observation #3



- The action of $\vec{f}$ along an axis of symmetry is simply to scale its input.

Observation #4



- The size of this scaling can be different for each axis.

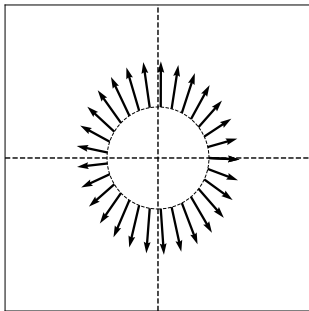
Main Idea

The **eigenvectors** of a symmetric linear transformation (matrix) are its axes of symmetry. The **eigenvalues** describe how much each axis of symmetry is scaled.

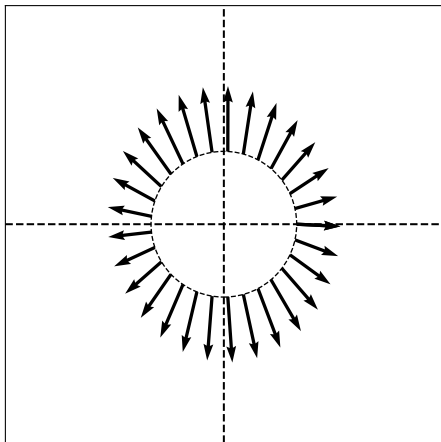
Diagonal Matrices

- If A is diagonal, its eigenvectors are simply the standard basis vectors.

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$$

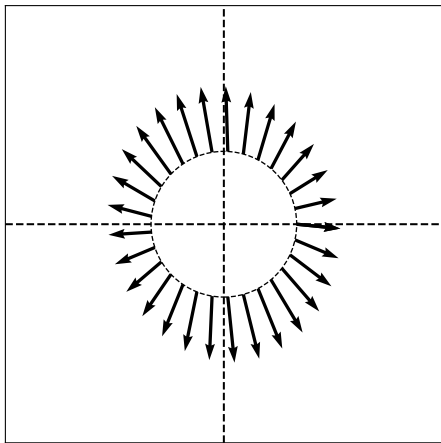


Off-diagonal elements



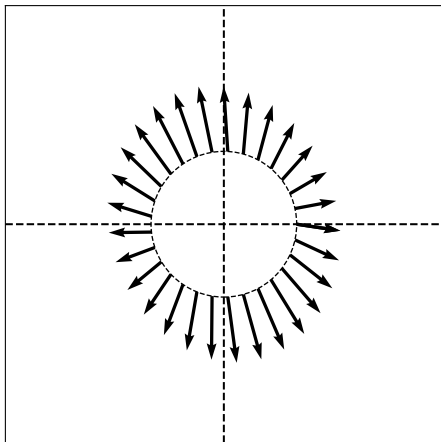
$$A = \begin{pmatrix} 2 & -0.1 \\ -0.1 & 5 \end{pmatrix}$$

Off-diagonal elements



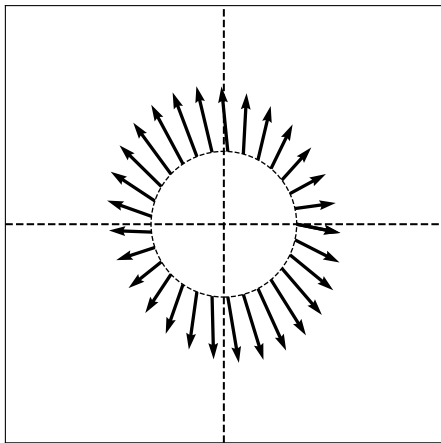
$$A = \begin{pmatrix} 2 & -0.2 \\ -0.2 & 5 \end{pmatrix}$$

Off-diagonal elements



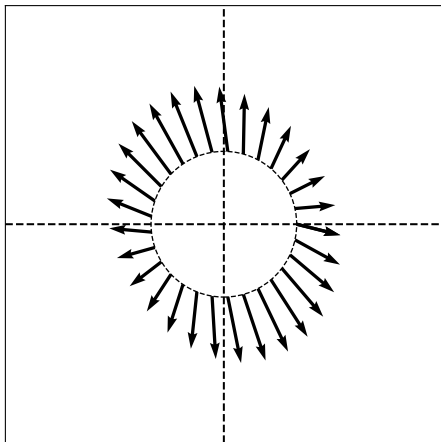
$$A = \begin{pmatrix} 2 & -0.3 \\ -0.3 & 5 \end{pmatrix}$$

Off-diagonal elements



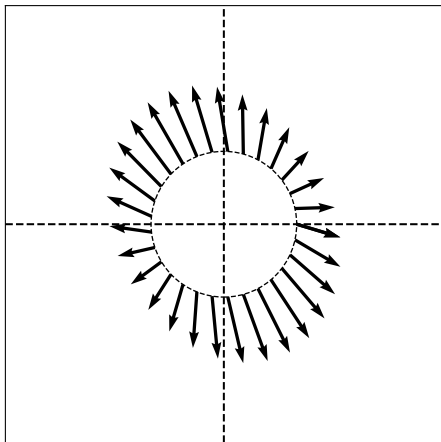
$$A = \begin{pmatrix} 2 & -0.4 \\ -0.4 & 5 \end{pmatrix}$$

Off-diagonal elements



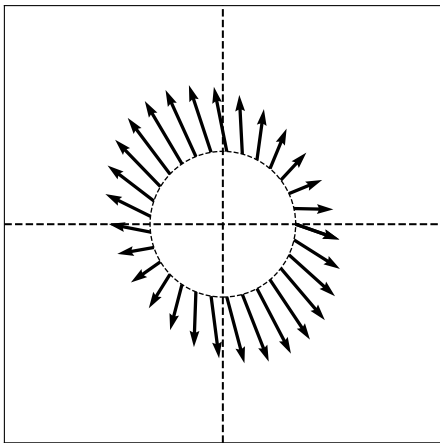
$$A = \begin{pmatrix} 2 & -0.5 \\ -0.5 & 5 \end{pmatrix}$$

Off-diagonal elements



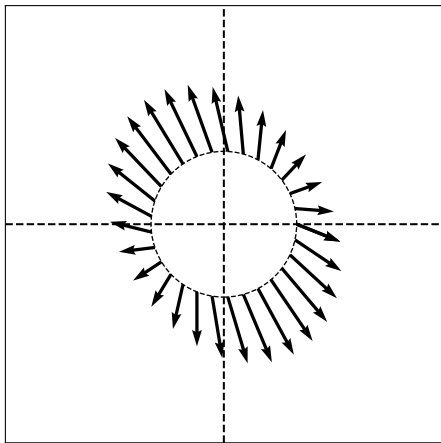
$$A = \begin{pmatrix} 2 & -0.6 \\ -0.6 & 5 \end{pmatrix}$$

Off-diagonal elements



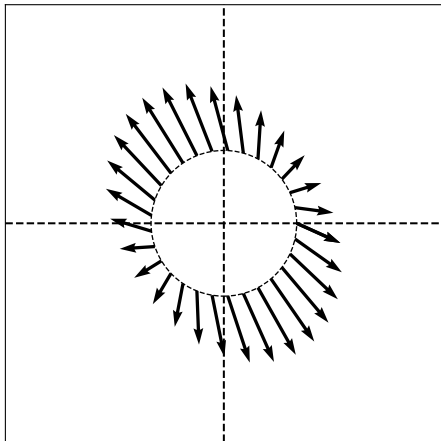
$$A = \begin{pmatrix} 2 & -0.7 \\ -0.7 & 5 \end{pmatrix}$$

Off-diagonal elements



$$A = \begin{pmatrix} 2 & -0.8 \\ -0.8 & 5 \end{pmatrix}$$

Off-diagonal elements



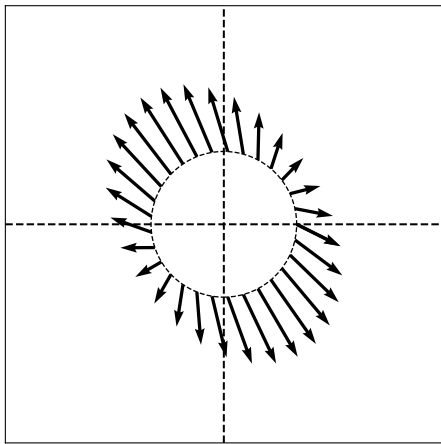
$$A = \begin{pmatrix} 2 & -0.9 \\ -0.9 & 5 \end{pmatrix}$$

Non-Diagonal Symmetric Matrices

- ▶ When a symmetric matrix is not diagonal, its eigenvectors are not the standard basis vectors.
- ▶ But they can be used to form an orthonormal basis!

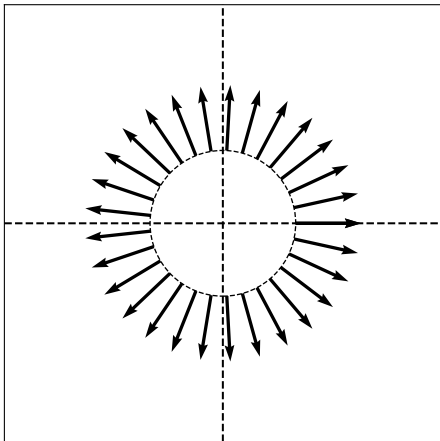
The Spectral Theorem⁵

- **Theorem:** Let A be an $n \times n$ symmetric matrix. Then there exist n eigenvectors of A which are all mutually orthogonal.



⁵for symmetric matrices

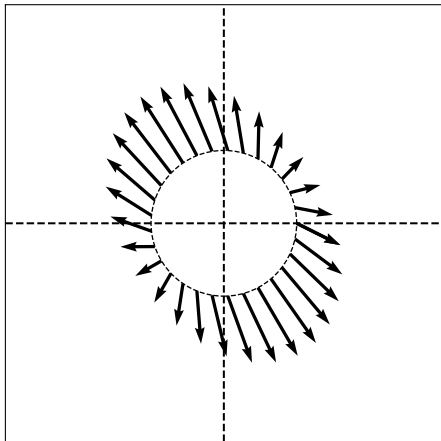
What about total symmetry?



- Every vector is an eigenvector.

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$$

Computing Eigenvectors



```
>> A = np.array([[2, -1], [-1, 3]])  
>> np.linalg.eigh(A)  
(array([1.38196601, 3.61803399]),  
 array([[-0.85065081, -0.52573111],  
         [-0.52573111,  0.85065081]]))
```